

Assignment mode:	Group of Four
Setter:	Ms. Narmada Gamage
Date set:	10/04/2026
Submission type	Presentation and Document
Assessment category	Group Assignment (Group registration link available in the course web)
Maximum marks:	50
Contribution to the final grade:	20%

Learning Objectives

In this assignment students will visualize and describe a dataset using **Power BI** software. At the end of the assignment students will be able to;

- Apply data visualization techniques
- Perform exploratory data analysis (EDA)
- Design dimensional data models (Star/Snowflake)
- Develop interactive dashboards using Power BI
- Apply basic predictive analytics
- Integrate data analytics with real-world ISPM projects.

Tasks

1) Dataset Selection & Preparation

In this assignment you should select a dataset directly related to their ISPM project and perform end-to-end data analytics .

- Select a real-world dataset (minimum 500 records) that includes at least two data sources requiring integration (e.g., sales, customer, and product data).
- *Note: You may use the same dataset selected for your ISPM project's novelty component. However, to avoid duplication, ensure that you provide the dataset name and link in the group registration form.*
- **If you are unable to find a suitable dataset that aligns with your ISPM project requirements, you may create a synthetic dataset that satisfies your project's novelty component.**
- Identify and document any data quality issues present in your dataset.
- Perform data cleaning and transformation using Power Query or any other suitable ETL method.

2) Data Modelling

- Create a star schema or snowflake schema for the prepared dataset.
- Clearly show fact and dimension tables, keys, and relationships using Power BI's modelling view.
- Explain your choice of schema and the type of relationships used.
- **Requirement: Model must reflect business process of your ISPM system**

3) Business Intelligence Dashboard

- Design an interactive Power BI dashboard that communicates actionable insights.
- Include a minimum of five visuals, covering:
 - Key Performance Indicators (KPIs)
 - Trend analysis
 - Geographic or category-based analysis
 - Predictive or forecast visuals
- **Requirement: Your dashboard must tell a business story covering problem / decision scenario, use visuals to explain insights and provide clear conclusions**

Ex: "Why customer churn is increasing", "Which products/services generate most revenue"

4) Prediction / Forecasting

- Apply basic predictive analysis, such as:
 - Forecasting trends
 - Demand prediction
 - Risk identification

5) Evaluation of Data Management Practices

- Reflect on limitations encountered and suggest improvements for future implementations.
- Individual components need to be described well.

Demonstration and viva

You must clearly demonstrate how your visualizations and findings were derived. All group members should have a thorough understanding of the entire assignment and be prepared to answer questions from any part.

Prepare a slide deck covering data issues, cleaning methods, schema design steps, predictions/forecasts, and evaluation with key findings and recommendations. Power BI visualizations should be explained using the dashboard, not included in the slides.

Presentations schedules will be available in the course web.

Document

You must include the following in your document

1. Introduction to the project and dataset
2. Identified data problems and cleaning methods
3. Schema diagram and relationship description
4. Power BI visualizations and interpretations
5. Predictions / Forecasting operations performed
6. Evaluation of data management practices
7. Findings and recommendations
8. Conclusion

Group registration deadline : 19th of April 2026

Assignment submission deadline : 1st of May 2026

Presentation date : 3rd of May 2026

Upload a **pdf** document and powerBI (.pbix) file to the course web link. Rename your file with the group ID.